

Biotvi-Blog-Rag-Agent

An intelligent AI agent built with n8n that answers questions about Bitovi's blog content using Retrieval-Augmented Generation (RAG).

Overview

This workflow creates an AI-powered chatbot that can answer questions like:

- "What is Bitovi's latest blog post about?"
- "Can you show me all Bitovi articles about DevOps?"
- "How many articles does Bitovi have about AI?"
- "What kind of tools does Bitovi recommend for E2E testing?"

Architecture

Data Ingestion Pipeline

- Automated scraping of all Bitovi blog articles
- Content extraction and cleaning
- Metadata preservation (titles, authors, dates, categories)
- Vector embeddings generation using Google Gemini
- Storage in Pinecone vector database

AI Agent with RAG

- Real-time question processing through chat interface
- Vector similarity search for relevant content retrieval
- Context-aware response generation with Google Gemini
- Source attribution with direct links to original articles

Technical Implementation

Prerequisites

- N8N self-hosted instance
- Google Gemini API key

- Pinecone API account
- Webhook URL for chat interface

Installation Steps

1. Import Workflow

```
# Import the provided JSON file into n8n
# File: bitovi-blog-ai-agent.json
```

2. Configure Credentials

- Add Google Gemini API credentials
- Add Pinecone API credentials
- Set up webhook for chat interface

3. Initialize Data Pipeline

- Activate the schedule trigger for automatic blog scraping
- Manual execution will index all existing articles
- Subsequent runs capture new content automatically

4. Test Chat Interface

- Send test queries through the webhook endpoint
- Verify responses include relevant content and source links

Workflow Components

Data Collection (Automated)

Schedule Trigger → Page Scraping → Content Extraction →
HTML Cleaning → Vector Embedding → Pinecone Storage

Question Answering (Real-time)

Chat Trigger → AI Agent → Vector Retrieval →
Response Generation → Source Attribution

Key Features

- **Comprehensive Coverage:** Ingests all Bitovi blog articles automatically
- **Smart Retrieval:** RAG pipeline with top-500 relevant results

- **Accurate Responses:** Grounded in actual blog content only
- **Source Tracking:** Every response includes original article links
- **Conversation Memory:** Maintains context across chat sessions
- **Error Handling:** Robust processing with failure recovery
- **Auto-Updates:** Scheduled content refresh for new articles

Response Format

Relevant information extracted from Bitovi blogs

Read more in <https://www.bitovi.com/blog/article-url>

Performance Metrics

- **Content Processing:** 800 items per batch
- **Retrieval Speed:** Sub-second response times
- **Coverage:** All historical and new blog posts
- **Accuracy:** Responses grounded only in blog content
- **Reliability:** Continuous operation with error recovery

Demonstration Capabilities

The AI agent successfully handles various query types:

Latest Content Queries

- Identifies most recent blog posts by publication date
- Provides summaries with direct links

Category-Based Searches

- Filters articles by topic (DevOps, AI, testing, etc.)
- Lists relevant articles with metadata

Quantitative Questions

- Counts articles by category or topic
- Provides statistical insights about blog content

Tool Recommendations

- Extracts specific tool mentions from articles
- Contextualizes recommendations with source attribution

